



Advancing the relevance of clinical trials for older patients

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Older patients represent the fastest growing patient group in clinical care. They are a heterogeneous group, for whom evidence for making treatment decisions is often scarce. Important domains for advancing the relevance of clinical trials for older patients are selection and inclusion of representative patients, choosing appropriate therapeutic interventions, and studying relevant outcomes. In the last decade, a myriad of publications has recommended multiple solutions that improve the relevance of trials for older people. Although uptake of these recommendations has been slow, there are now some excellent and successful implementation approaches that show that it is possible to design and execute trials that are inclusive of older patients, address relevant outcomes, and allow for real-world comparisons of intervention effectiveness. Observational studies based on different data sources add useful complementary information. This narrative review aims to synthesise best practices and successful strategies to improve the relevance of clinical trials for older patients.

Introduction

Older patients represent the fastest growing subgroup in clinical care. WHO estimates that the number of people aged 65 years or older will double worldwide between 2020 and 2050,¹ and for those aged 80 years or older, the number will triple. As people age, they are more likely to experience chronic diseases, physical and mental impairments, and social issues.² Additionally, their needs, preferences, and priorities might change. Moreover, the ageing process is highly heterogeneous, and multimorbidity, frailty, cognitive impairment, and disability are important determinants of diversity in clinical presentations. Frailty is characterised by decreased functional reserves and increased vulnerability to stress

and acute adverse events and is associated with worse clinical outcomes when the patient is subjected to a harmful stimulus, such as interventional procedures or drug toxicity.³ These challenging factors can complicate clinical decision making and participation in clinical trials, and can influence response to therapeutic interventions. Clinical trial inclusion and exclusion criteria and other barriers can lead to insufficient accrual of the older and frail subpopulations.⁴ Due to this under-representation, relevant questions for the patient, such as whether they will be able to tolerate the treatment and the chances of remaining independent after the disease and its treatment, cannot be answered with a high level of evidence. The same counts for clinicians, who should not prescribe drugs to this population without knowing if they will be effective and safe. The panel illustrates how the individual situation of an older woman with chronic kidney disease affects the applicability of trial results to her specific situation and questions she might have about participation.

Physicians and patients try to avoid both overtreatment and undertreatment as much as possible, since both negatively affect cost-effectiveness, treatment effectiveness, and quality of life (QoL).⁵ Extrapolation of clinical trial results to populations not studied nor eligible for inclusion in clinical trials can pose risks.⁶ Nearly all (99%) of new drugs approved in Europe and the USA between 2012 and 2023 were approved for patients who were less fit compared with the trial populations in which drug efficacy and tolerability were tested.⁶ Additionally, only half of the new US Food and Drug Administration (FDA) cancer drug indications between 2015 and 2021 were approved based on trials with adequate representation of older patients, defined as a participation to prevalence ratio of at least 0.8.^{7,8}

There are many barriers to overcome to advance the relevance of clinical trials for older patients, such as not selecting and including representative patients, using inappropriate interventions, and studying irrelevant outcomes (panel). In the last decade, many articles have been published that suggest a range of solutions.^{9,10} Although uptake of these suggestions has been slow, there are now some excellent implementation approaches demonstrating that it is possible to design and execute

Search strategy and selection criteria

To identify barriers and potential solutions related to trial design and execution for older patients, we performed searches in MEDLINE, PubMed, and searched references in relevant articles using the keywords “older patients” and “trial design” published between Jan 1, 1990 and Dec 31, 2025. Systematic reviews and meta-analyses were also specifically consulted for this purpose. Subsequently, we used expert knowledge to identify best-practice examples of trials that had resolved one or more of the identified barriers. We aimed to provide examples of trials in different fields of medicine and decided to use one trial example that successfully addressed the problem for each identified barrier, based on consensus among the authors. The focus was on drug trials, but we also allowed other domains if they were fit for purpose. We discuss best-practice examples divided into two main categories: trials designed for all ages (referred to as all-age trials), and trials designed specifically for older patients, in the absence or presence of frailty. Barriers and solutions in trial designs are structured in three main subcategories: patient inclusion, selection, and characterisation; interventions; and outcomes. Additionally, we discuss different study designs that resolve barriers in clinical trials and implementation strategies.

trials that are inclusive of older patients, address relevant outcomes, and allow for comparisons of intervention effectiveness between the trial and real-world populations. This narrative review aims to synthesise best practice and successful strategies to improve the relevance of clinical trials for older patients.

Improvement of clinical trials for older people in all-age trials

Patient inclusion, selection, and characterisation

A 2025 survey of clinical trials registered at clinicaltrials.gov showed that more than 75% of phase 2 and 3 interventional trials completed between 2020 and 2024 allowed inclusion of older adults (aged ≥ 80 years).¹¹ Thus, upper age limits are no longer the most important barrier to the inclusion of older patients. Despite this development, a mismatch remains between the prevalence of various diseases in older patients and their participation rate in clinical trials. Comorbidity, polypharmacy, and nursing-home residence are common criteria leading to the exclusion of older patients.¹² However, with broad inclusion criteria, recruitment of a representative study population is feasible, as shown in the IST-3 trial, which investigated the use of alteplase in acute stroke.¹³ The trial had broad inclusion criteria and 53% of patients enrolled were older than 80 years. To ensure sufficient enrolment of older patients, careful assessment is needed of the rationale for any exclusion criterion, as well as the effect each will have on the external validity of the trial.

Nonetheless, achieving adequate representation remains difficult due to several barriers (table).¹² The table describes these barriers and reasons behind the scarce data in older populations, as well as possible solutions. One perceived barrier is the willingness of older patients to participate in clinical trials. In general, older patients appear as willing as their younger counterparts,⁵⁶ but understanding trial procedures could pose a barrier.⁹ Incorporating patient education regarding informed consent or adapted study procedures (such as audiovisual information sources^{57,58} or increased contact with a research nurse⁵⁹) into the trial protocol might increase trial enrolment. A study on percutaneous nephrolithotomy showed that using video-assisted education about the procedure resulted in significantly greater comprehension compared with written information only.²⁵ Although the study did not include many older adults, this type of patient education could improve understanding of informed consent procedures and thus, increase patient enrolment.

A lack of motivation from physicians to include older patients in clinical trials is another barrier. Integrating research staff training into the trial protocol and budget can help alleviate this issue. In the EnTraP study, an educational intervention to trainee primary investigators and positive reinforcement of trial recruitment resulted in a 23% increase in enrolment of patients of any age

Panel: Consequences of current trial design for clinical decision making and trial inclusion in a real patient

Case

Mrs A is aged 78 years, with newly diagnosed stage III breast cancer, and chemotherapy is considered. She lives with her husband; he has heart failure and she helps him with bathing and his medications. She values her independence, enjoys working in her garden, and spending time with her family. She is discussing treatment options and potential trial participation with her treating physician and would like to know more about the benefits and risks.

Findings from geriatric assessment

- Mild frailty (slowed gait, reduced stamina, but independent in daily activities)
- Mild forgetfulness affecting planning and organisation
- Social support: weekly visits from children and grandchildren; neighbour assists with transport as she does not have a driver's licence
- Comorbidities: type 2 diabetes, chronic kidney disease stage III

Dilemmas in decision making with currently available evidence

Patient characteristics

Evidence from selected trial populations regarding treatment benefit might not apply to real-world patient populations who did not fulfil eligibility criteria, making it difficult to adequately inform this patient.

Intervention

Mrs A is concerned that treatment side-effects will affect her ability to take care of her husband; it is unknown what the treatment tolerability is in older patients with comorbidities.

Outcomes

Mrs A prioritises maintaining her independence and cognitive function; evidence generated by conventional trial endpoints does not provide information about these outcomes.

Potential barriers to trial inclusion

Patient characteristics

Age, comorbidities (such as diabetes and chronic kidney disease), and mild cognitive impairment might lead to exclusion under strict trial criteria.

Intervention

Mrs A prefers the treatment arm that involves the least treatment burden and number of hospital visits, so she does not need to ask her neighbour for transportation or leave her husband alone at home, but randomisation means it is uncertain which treatment arm she will be placed in.

Outcomes

Competing risk makes outcomes harder to specify; collecting outcomes other than mortality (eg, functional decline or maintenance of independence) is more complicated

compared with usual recruitment practices.²⁷ Active stakeholder engagement in all phases of clinical trial design can also boost patient and physician motivation and trial feasibility. The TrACER trial²⁰ used an external stakeholder advisory group, including patients, payers, providers, and various experts, which provided input on recruitment strategies, trial procedures, endpoints, and patient education. A meta-analysis showed that involving patients and the public in co-developing trial designs improves patient enrolment in randomised controlled trials (RCTs).⁶⁰ When specifically aiming to enhance representation of older patients, including a geriatrician on the study team is essential,³⁶ as was done in the

	Successful trial example	Strengths of study design
Patient inclusion, selection, and characterisation		
Exclusion of older patients based on age or strict inclusion criteria ¹⁴		
Adopt broad inclusion criteria and critically assess the necessity of each potential exclusion criterion ^{2,10,15}	IST-3 ¹⁹ showed that thrombolysis within 6 h improved overall survival in patients with ischaemic stroke	Wide inclusion criteria; included 53% of patients aged >80 years
Develop trials specifically for older, frail, or multimorbid patients by tailoring exclusion criteria ^{10,15}	GAP70+ ¹⁶ showed that a geriatric assessment with targeted management recommendations significantly reduced grade 3–5 chemotherapy toxicity within 3 months among patients >70 years with advanced cancer, compared with usual care	Patients required at inclusion to have at least one impaired geriatric domain, such as poor physical performance, comorbidities, malnutrition, or functional or cognitive deficits—effectively targeting older adults who are vulnerable or frail
Exclusion from trial based on concerns or lack of clarity regarding cognitive ability to consent		
Implement procedures to deal with concerns and issues regarding cognition and consent during the inclusion phase ^{2,17}	MAP ¹⁸ is an ongoing trial that investigates the effect of metformin to prevent cognitive decline in patients with mild cognitive impairment without diabetes	Tool for assessing capacity to consent
Consider consent by proxy when capacity to consent is lacking	In a trial with rivastigmine, it was shown that this drug led to moderate improvement in dementia associated with Parkinson's disease ¹⁹	Allowed for consent by a family member when cognitive impairment affected capacity to consent
Failure to sufficiently recruit older patients, especially those living with frailty, resulting in an unrepresentative study population or not achieving target accrual or primary endpoint		
Integrate active stakeholder (eg, health-care providers, patients, caregivers, and public) engagement and co-design ^{2,10,15}	TrACER ²⁰ is an ongoing pragmatic trial assessing the effectiveness of a guideline-based colony stimulating factor standing order intervention in patients receiving chemotherapy	External stakeholder advisory group with patients, payors, pharmacists, guideline experts, providers, and an ethicist, which informs and engages in trial design, dissemination, and implementation
Develop a protocol for targeted recruitment of older or frail patients ^{2,15,21}	SPRINT ^{22,23} showed that a systolic blood pressure 120 mm Hg compared with 140 mm Hg resulted in lower rates of fatal and non-fatal major cardiovascular events	Protocol for targeted enrolment of specific subgroups, including patients aged ≥75 years; met recruitment goals within 28 months; included a frailty index at baseline
Use multiple channels of communication to recruit; target caregivers in addition to patients ²	MTOPS ²⁴ showed that finasteride treatment compared with placebo led to an increase in the percentage of patients who did not have progression of benign prostatic hyperplasia	Used several recruitment strategies including emails, newspapers, medical sources, and radio advertisements; also published the most effective recruitment strategies (newspaper and mail)
Little (time for) patient education to motivate them for trial participation and address concerns		
Integrate additional patient education strategies in the trial protocol	Video-assisted education in informed consent procedures resulted in greater comprehension in patients who underwent percutaneous nephrolithotomy ²⁵	Used video-assisted education in providing informed consent for the procedure, and patients were randomly assigned to this procedure or written consent only
No physician education or limited motivation for including older patients in trials ²⁶		
Include training of research staff into the trial protocol and trial budget ^{2,15}	EnTraP ²⁷ showed that an education and support programme targeted at surgical trainee principal investigators was effective in increasing recruitment of a clinical trial	Patients randomly assigned to the educational programme or standard practice to show approach effectiveness in a large clinical trial
Allow for stratification in subgroups based on physician-assessment or frailty, thereby adapting randomisation levels of treatment intensity to frailty ¹⁰	GO2 ²⁸ showed that upfront dose-reduced chemotherapy resulted in similar overall survival in older patients with advanced upper-gastrointestinal cancer	Patients stratified by frailty, those considered unfit for full-dose treatment were randomly assigned (60% chemotherapy vs best supportive care) in a separate randomisation arm
Trial-related procedures required from the patient or caregivers are too burdensome or too complex to understand ^{10,15}		
Use short, simple consent forms and limit trial-related burden as much as possible; offer alternatives to full clinic visits (such as telephone consultations) ^{2,15,29}	PREVENTABLE ³⁰ is an ongoing trial investigating the effect of a preventive statin on survival free of new dementia or persistent disability in older community-dwelling adults ≥75 years	A pragmatic design with flexible inclusion and follow-up regimens
Design pragmatic, flexible, and decentralised trials (eg, by integrating home visits) to decrease trial burden for the patient and their caregivers ^{2,10,31}	IEMO 80-plus ³² showed that treatment with levothyroxine in patients ≥80 years with subclinical hypothyroidism did not improve hypothyroid symptoms or fatigue	Offered to perform all study related procedures at the place of residence of the participants
Little or no available data to allow for comparison with real-life population		
Use a core (geriatric) dataset to characterise the full trial population or embed this in a companion study specifically for older patients ^{2,10,31,33}	HYVET ³⁴ showed that antihypertensive treatment in older patients (>80 years) resulted in lower blood pressure and a reduction in stroke and cardiovascular events	Used core geriatric dataset at baseline to characterise the population
Allow for pragmatic execution of geriatric or frailty assessment to facilitate alignment with existing care	The GERONTE trial ³⁵ evaluates the benefit of a new care pathway for older patients with multimorbidity and cancer	Only standardises which geriatric domains are assessed and how frailty results are reported, while leaving the choice of assessment methods to the investigator
Insufficient input of geriatricians in study design		
Integrate geriatricians in the study design and conduct	FAST-MAG ³⁶ showed that paramedic infusion initiation with magnesium in patients with acute stroke did not improve clinical outcome	Geriatricians were included in the research team and assisted in study procedures and interpretation; included 550 patients aged >80 years of whom 60 were aged >90 years

(Table continues on next page)

	Successful trial example	Strengths of study design
(Continued from previous page)		
Intervention		
Treatment burden of the trial (or one of the trial arms) is too intense for older patients to tolerate ^{10,15,37}		
Use adapted treatment intensity in the experimental arm, or in both arms (in a trial specific for older patients) ³⁰	MRC FOCUS2 ³⁸ was a 2 × 2 factorial trial comparing 5-fluorouracil vs capecitabine chemotherapy, with or without oxaliplatin; the trial showed that a combination with oxaliplatin was preferable, although the primary endpoint was not met	Targeted frail patients and used adapted chemotherapy regimens with reduced doses of all regimens in all arms, which made the trial feasible for frail patients who could not tolerate full-dose treatment
Have an adapted intervention with a run-in (ie, test) period in a small number of patients to assess safety before enrolling a larger cohort of patients	TRADE-hypo ³⁹ investigated safety and efficacy of durvalumab with either hypothoracic or conventionally fractionated thoracic radiotherapy	Had a run-in period to assess safety; after inclusion of 38 older patients, two major adverse events occurred in the intervention arm, after which a recruitment stop in this arm was performed (per-protocol); inclusion in the control arm continues
Gradual dose up-titration of potentially toxic drugs to avoid excessive toxicity at the start	DESIREE ⁴⁰ showed that stepwise dose escalation of everolimus resulted in a reduced incidence of stomatitis in patients with metastatic hormone receptor-positive breast cancer	Patients were randomly assigned to standard treatment or a lower starting dose with stepwise increase in dose
Design a parallel trial for patients unfit for the primary trial	FOXOTROT2 and FOXOTROT3 ^{41,42} are ongoing trials in patients with locally advanced colon cancer testing different neoadjuvant regimens	Patients who are unfit for the most intensive neoadjuvant treatment (in FOXOTROT3) can be enrolled in FOXOTROT2, which uses a milder regimen
Include an observational arm for patients unfit to be randomly assigned ^{10,31}	DOSAGE ⁴³ is an ongoing trial investigating dose-reduced chemotherapy in older patients with metastasised colorectal cancer	Includes patients who do not want to be randomly assigned (or are deemed unfit for random assignment by their clinician) in an observational study arm
Optimise treatment, support, and management of comorbidities and geriatric impairments to increase trial feasibility	PERFORM-TAVR ⁴⁴ is an ongoing trial in patients who will undergo transcatheter aortic valve replacement; the trial will investigate a programme including nutritional and exercise interventions	Examines the synergistic effects of exercise and protein supplementation to counteract frailty and improve patient-centred outcomes following transcatheter aortic valve replacement
Outcome		
Relevant outcomes for older patients are not included or not reported ^{7,45}		
Choose endpoints that are meaningful to older people, such as functional autonomy, cognition, depression, and nutrition ⁴⁶	DANTON ⁴⁷ showed that discontinuing antihypertensive treatment in nursing-home residents with dementia resulted in a non-significant increase in serious adverse events	Used the Neuropsychiatric Inventory Nursing Home Scale and quality of life as co-primary endpoints
Use patient-reported outcomes, such as health-related quality of life, patient's symptoms, or satisfaction as primary endpoints	TRUST ⁴⁸ showed that treatment with levothyroxine did not result in clinical benefit in older patients with subclinical hypothyroidism	Used two primary endpoints that were relevant for patients in this context (change in hypothyroid symptoms score and tiredness score)
Assess benefit of interventions by combining patient-reported outcomes with other endpoints to form co-primary or composite endpoints	ASPRE ⁴⁹ assessed the effect of aspirin on disability-free survival	Used a primary composite endpoint of death, dementia, or persistent physical disability
Analyse composite endpoints with generalised pairwise comparisons, allowing for simultaneous assessment of multiple endpoints and prioritisation	SHAPERS ⁵⁰ is an ongoing trial investigating whether neoadjuvant short-course radiotherapy is a better trade-off between safety and efficacy than total neoadjuvant therapy in patients with locally advanced rectal cancer	Uses a multicomponent measure of treatment effect based on generalised pairwise comparisons as a primary endpoint; this measure includes overall survival at 3 years, progression-free survival, neuropathy, and grade ≥3 toxicity
Use hierarchical endpoints (ie, first confirm non-inferiority of lower intensity treatment, and if confirmed evaluate patient-reported outcomes to establish the best treatment approach)	EMPULSE ⁵¹ showed that empagliflozin in patients hospitalised with heart failure improved clinical benefit	Used the hierarchical endpoint clinical benefit, defined as death from any cause, number of heart failure events, and time to first heart failure event, or a 5-point or greater difference in change from baseline in the Kansas City Cardiomyopathy Questionnaire Total Symptom Score at 90 days
Competing risks are insufficiently considered in the trial design		
Incorporate competing risk in the trial design and report appropriate statistical analyses ⁵²	MOSCA-FRAIL ⁵³ showed that an invasive treatment strategy did not improve clinical outcomes in older, frail patients with non-ST-segment elevation acute myocardial infarction	Used a competing risk event analysis for cardiac death with non-cardiac death as a competing event as a co-primary composite endpoint
No age-specific analyses or consideration for heterogeneity in the study cohort		
Routinely plan and execute analyses for subgroups based on age, functioning, and comorbidity	SENIOR-RITA ⁵⁴ showed that an invasive treatment with revascularisation in older patients with non-ST-segment elevation acute myocardial infarction did not improve clinical outcome	The trial included prespecified subgroup analyses including different levels of physical frailty, comorbidity, and cognitive impairment
Insufficient number of older people available for subgroup analyses		
Perform pooled analyses or meta-analyses of different trials in a similar population	A pooled analysis of five adjuvant chemotherapy trials in older patients with breast cancer showed that taxane-based adjuvant chemotherapy was superior to alternative regimens, although with higher toxicity ⁵⁵	Pooled patient-level data from older patients who were enrolled in five previous phase 3 trials
Table: Barriers and strategies to improve clinical trial relevance for older adults		

FAST-MAG trial.³⁶ In France, geriatric oncology coordination units have been created, not only for improving patient care but also enhancing collaboration in the inclusion process for cancer trials.⁶¹

Specific recruitment strategies can further boost the representation of older patients. The MTOPS trial,²⁴ which evaluated treatment for benign prostatic hyperplasia, used a broad array of promotion strategies before the start of recruitment, including mass mailing, radio, television, newspapers, screening of medical records, and presentations to physicians. Additionally, men who were screened were given brochures to disseminate to friends, relatives, and co-workers. Another way of enhancing feasibility of older patient participation is decreasing the burden of trial-related activities. A study assessing preferences for participation of patients of all ages with type 2 diabetes in clinical trials, with different options for location and type of contact with the study team, found that decentralised trials involving video contact with the study team had the highest participation probability.⁶²

In addition to improving recruitment, clinical evidence can be enhanced by reporting more detailed information regarding the health status of participating patients, including aspects that are heterogeneous among older trial participants (ie, the degree of frailty, level of care dependence, functional status, or presence of cognitive impairment). Although these elements are increasingly being included,⁶³ they are rarely reported, even in trials specifically designed for older patients.⁶⁴ Accurate and complete reporting allows for real-world comparisons of results and estimation of benefits and tolerability.^{65,66} Tools can be developed specifically for a trial, such as in the SPRINT hypertension trial,²² which developed a geriatric assessment and frailty instrument including cognition, mood, and functioning parameters that was administered to all patients. Alternatively, various recommendations regarding the content of a frailty assessment are available.^{65,66} For example, the UK Geriatric Medicine Core Dataset Extended Working Group proposed six core domains including demographics, specified morbidities, functional ability, Clinical Frailty Scale, cognition, and patient-reported outcome (PRO) measures.⁶⁷ The Frailty Outcomes Consensus Project study group⁶⁸ was developed to reach consensus from the international frailty community on a set of common data elements and core outcomes for frailty, and concluded that medications, mobility, activities of daily living, comorbidities, cognitive impairment, informal care and support, cumulative deficits leading to frailty, and QoL were the primary data elements to be included. To increase the practical feasibility of centres participating in a multicentre trial, an option could be to limit protocolisation to the domains and reporting of the frailty assessment results, leaving the responsibility of the assessment methodology itself to the individual centres,⁶⁹ as seen in the GERONTE trial.³⁵ This way, centres do not have to change care pathways already in place in order to participate in the trial, making

trial participation less burdensome for each centre.^{33,70} In many all-age trials, it might not be possible to collect a full geriatric assessment for the older group of enrolled patients for feasibility reasons. In that case, a minimum core dataset can be a good option, such as proposed by the European Organisation for Research and Treatment of Cancer.⁷¹

Interventions

Inevitably, therapeutic interventions tailored to fit patients in all-age trials will not always be suitable for older patients, especially for older adults with decreased physiological reserves or comorbidities. Developing a trial specifically for vulnerable patients could be the most feasible option, as discussed in the next section. However, there are possibilities to enhance all-age studies to improve evidence for older patients. Parallel trials (ie, enrolling older adults in a separate cohort) and embedded studies (ie, a separate study collecting additional assessments or biomarkers in a patient subset with specific characteristics) can facilitate the capture of more ageing-specific measures to characterise an older population or evaluate the effects of the intervention on an endpoint relevant to older adults.³¹ For example, FOxTROt3⁴¹ investigated the role of intensification of neoadjuvant chemotherapy in fit patients with locally advanced colon cancer. If a patient was unsuitable for intensified treatment, they were considered for FOxTROt2,⁴² which addressed de-intensified neoadjuvant chemotherapy in locally advanced colon cancer, across the full range of older adults, including those with frailty.

Outcomes

Primary endpoints in trials tend to be disease-centred, such as event-free or overall survival.⁷² However, patient-centred endpoints, such as independence, physical function, or QoL, are at least as relevant as disease-centred endpoints for older patients who present more frequently with comorbidities, deaths due to other health problems, and treatment-related adverse events and toxicity.⁷³ In addition, the expectations and preferences of older patients regarding trial outcomes might differ from those of younger ones. The International Consortium for Health Outcomes Measurement has developed an outcome set specific to older patients incorporating the domains that matter most to older patients;⁷⁴ the consortium also presents a range of disease-specific outcomes that might be useful in clinical trial design.⁷⁵

It is crucial to assess patients' feelings and experiences when evaluating new treatments in clinical trials.⁷⁶ PROs are a measure of a patient's health condition, symptoms, or treatment effects as reported by the patient, without interpretation by clinicians or others. PROs can include health-related quality of life (QoL), symptoms, or satisfaction. Several tools can be used to measure these domains, but there is a need to define minimum clinically important differences in older patients when

using such tools as endpoints in trials.⁴⁶ Selecting the endpoints and prioritising them as primary and secondary outcomes through a collaborative process involving investigators and patient partners or carers helps to ensure that the choices made reflect what patients consider most important. Older patients can participate in research and its design, implementation, and dissemination, similar to their younger counterparts.

A relevant point to add is that the information collected during clinical trials to monitor safety of drug treatment is often inadequate to identify important side effects for older adults, including cognitive or functional decline, falls, and delirium.⁷⁷ It should be acknowledged that classical trial endpoints such as overall or event-free survival maintain their relevance for older persons as well and should not be completely replaced by PROs.

Clinical trials specifically developed for older patients (older age-specific trials)

Many of the issues and strategies discussed in the section on all-age clinical trials are also relevant for clinical trials developed specifically for older adults. Although older age-specific trials can provide essential evidence, they also have their own challenges and corresponding solutions.

Patient inclusion, selection, and characterisation

Older age-specific trials allow for making inclusion and exclusion criteria more flexible and representative for older people, resulting in higher accrual potential, especially for those living with frailty.⁷⁸ The DANTON trial⁴⁷ investigated the effect of discontinuation of antihypertensive medication on neuropsychiatric symptoms in patients with moderate-to-severe dementia living in nursing homes. The study had wide inclusion criteria allowing for recruitment of patients with moderate-to-severe dementia and frailty. Inclusion procedures were pragmatic: patients were identified through screening by the treating physician, and informed consent was provided by the patient's legal representative, thereby making inclusion less burdensome for patients. The study closed prematurely, not because of poor recruitment (the study recruited at the target rate despite the COVID-19 pandemic), but because the interim analyses showed negative outcomes in the intervention arm, thereby providing highly relevant information for this vulnerable group of patients. Designing pragmatic, flexible, and decentralised trials can help to decrease the trial burden for the patient and their caregivers. The ongoing PREVENTABLE trial on the effect of preventive statin on survival free of dementia or persistent disability used flexible inclusion and follow-up regimens, which were similar to usual care, to increase trial feasibility.³⁰ Home visits for inclusion and retention can also help, especially for those who are frail or have transport problems, as shown in the IEMO 80-plus trial.³²

Cognitive impairment requires special attention in trials designed for older patients. Its presence is often

unrecognised; in one study of more than 200 individuals with no previous diagnosis of cognitive impairment, more than 60% had mild cognitive impairment and more than 12% had dementia when formally assessed.⁷⁹ As the prevalence of cognitive impairment increases with age, it is imperative that participants with cognitive impairment are identified through screening measures, so that more attention can be provided (ie, diagnostic procedures for cognition and evaluation of risks and benefits of trial participation). The GAP70+ study used a screening process to evaluate potential cognitive impairment for older adults with cancer; if a patient screened positive for impairment, the clinician was notified in both the intervention and usual-care arms.¹⁶

Another subgroup that deserves specific attention is nursing-home residents. These patients are often excluded from trial participation due to their place of residence, irrespective of their remaining life expectancy, level of disability, or care needs. Careful consideration of the necessity for such upfront exclusion could help improve trial representativeness. Alternatively, trials focusing specifically on nursing home residents—such as the DANTON trial described above⁴⁷—might be helpful in increasing evidence for this population. These trials have their own execution challenges. Through the Older Americans Independence Center's AGING Initiative, a multidisciplinary stakeholder advisory panel developed recommendations to improve the capability of the clinical research enterprise in nursing homes, addressing coordination, leadership, stakeholder engagement, funding, and regulations.⁸⁰ As with all-age trials, trials developed for older patients should characterise the study population adequately by using a geriatric assessment or frailty dataset as described in the previous section.

Interventions

Several trial designs can facilitate an increased understanding of efficacy and tolerability of therapeutic interventions in older adults with varying fitness levels. By including multiple stratification arms based on frailty or physician judgement, trials can test several regimens in different subgroups. For example, the currently ongoing DOSAGE trial⁴³ investigates palliative chemotherapy regimens in older patients with metastatic colorectal cancer, and stratifies patients based on frailty. Patients with frailty are randomly assigned to full-dose or reduced dose monotherapy, whereas fit patients are randomly assigned to full-dose or reduced dose polychemotherapy; an additional observational arm is available for patients unwilling to be randomly assigned, making it more likely that patients with severe frailty are also included in the study. Of note, as well as a method of selecting and characterising the patient population, frailty assessment could also be included as an additional treatment goal to achieve the best clinical outcome after therapeutic intervention. A good example is the transcatheter aortic valve replacement protocol in the

PERFORM-TAVR trial,⁴⁴ which examines the synergistic effects of exercise and protein supplementation to counteract frailty and improve patient-centred outcomes following aortic valve replacement.

Incorporating a run-in period, in which a new intervention is tested for safety in a small group of patients, can be another alternative to improve enrolment of older adults with frailty, as was shown in the TRADE-hypo trial.⁵¹ In recent years, de-escalation trials in which treatments are given in a lower dose or even omitted have gained popularity, especially in oncology. Often, these trials use a non-inferiority design, where the standard of care is being compared with de-escalation (such as the GO2 trial²⁸). A limitation of this design is that patients must be able to tolerate the full dose, as they could be randomly assigned to this arm. An alternative is to have adapted treatment regimens in both intervention and control arms to compare different adapted regimens in frail populations, as was nicely shown in the FOCUS2 trial.³⁸ This design can provide insight into differences between adapted regimens in frail populations but obviously does not allow for comparison with standard treatment regimens.

In addition to the specific interventions under investigation, trials can also incorporate specific procedures to accommodate for the characteristics of older patients and improve retention of these patients for the duration of the trial. In a trial using invasive right-sided heart catheterisation in healthy older patients,⁸² a research nurse was closely linked to the patient to provide additional explanations, answer any questions throughout the study duration, provide reassurance by being physically present during proceedings, and arrange transportation to and from the clinic if needed. Retention can also be improved by providing reimbursement for expenses, regular information about the progress of the study, expression of appreciation by the staff, and having a low staff turnover.^{82,83}

Outcomes

Like in all-ages trials, the endpoints of trials for older people need to be meaningful for them, but the priorities of older people might switch to more functional and quality-related outcomes.^{84,85} PROs should be used more often as primary endpoints. The DANTON trial⁴⁷ used neuropsychological symptoms and QoL in nursing-home residents with dementia as primary endpoints. Another example is the TRUST trial,⁴⁸ which investigated whether levothyroxine treatment benefits older adults with subclinical hypothyroidism. The primary endpoints were changes in hypothyroid symptoms at 1 year and tiredness scores on a thyroid-related QoL questionnaire. PROs can be combined with other endpoints to assess the benefit of treatment. Co-primary endpoints include two or more distinct outcomes that must all be met to conclude the trial's success; they offer the advantage of providing a comprehensive assessment of treatment effects across multiple outcomes, reducing the risk of erroneous efficacy

conclusions. However, this approach increases complexity, can increase sample size, and results in stringent success criteria since all co-primary endpoints need to show significant results for the outcome to be considered positive. The aforementioned TRUST trial⁴⁸ used two PRO measures of QoL as co-primary endpoints. Composite endpoints are another way of combining multiple endpoints by creating a single measure in which the occurrence of any one of the component events counts as the endpoint. The ASPREE trial,⁴⁹ assessing the effect of aspirin on disability-free survival, used a primary composite endpoint of death, dementia, or persistent physical disability. This approach increases event rates and trial efficiency but can pose challenges in interpretation due to varying clinical importance of components. Additionally, negative and positive findings (ie, more frequent adverse events but better survival) might neutralise each other. As was done in the ASPREE trial, it remains important to report the outcomes separately as well.

There is a need to develop valid composite endpoints that integrate treatment efficacy, tolerability, and QoL. Several new endpoints have been developed combining multiple primary endpoints in older patients. The modified Rankin scale captures multiple levels of functional decline and mortality in one score and is the most widely employed outcome scale in acute stroke trials.^{86,87} The SHAPERS trial⁵⁰ studied older patients with acute promyelocytic leukaemia and introduced net treatment benefit, a composite endpoint, using generalised pairwise comparisons of five disease-centred and patient-centred outcomes to confirm a favourable benefit or risk of a less intensive treatment regimen. This statistical method allows for simultaneous assessment of efficacy and safety, prioritising the efficacy.⁸⁸ Hierarchical endpoints can also be used in the older population. The EMPULSE trial⁵¹ used this approach and assessed clinical benefit of empagliflozin in patients who were in hospital for acute heart failure. Clinical benefit was expressed as a win ratio between the two treatment arms. Death was assessed first, with more deaths or earlier deaths considered as poorer outcomes. If no winner was evident based on this assessment, heart failure-related events were compared. The third level was the time to first heart failure-related event, and if that still did not show a significant difference, the final comparison was based on a symptom score. QoL is arguably the most relevant outcome for older people when considering therapeutic interventions. A commonly encountered problem in trials is that repeated measurement of QoL is nearly impossible in all patients because of dropout for premature death or withdrawal because of excessive adverse treatment-related events. Bias might be introduced when QoL is only measured in those patients who remain able to complete assessments during follow-up. A recent manuscript summarised different statistical techniques to overcome these

issues.⁸⁹ Using a composite endpoint (combining QoL and death) can be appropriate when poor QoL is considered equally as important as reduced life span.⁸⁹ Finally, competing risk of death can substantially influence endpoint assessment in older people, who are at higher risk for this factor. In the MOSCA-FRIL study,⁹⁰ advanced statistical methods were developed to analyse survival data in the presence of competing risks, such as cumulative incidence functions, to estimate the incidence of the outcome over time in the presence of competing risks.⁵²

When older adults are enrolled in clinical trials, standardised reporting³³ of outcome in relation to age can increase knowledge about efficacy and tolerability of treatments. Subgroup analyses, using stratification based on age or on geriatric parameters, provide information that is essential for well informed shared decision making in older patients. For example, in the SENIOR RITA trial⁵⁴ on non-ST elevation myocardial infarction in which patients were assigned to a conservative or invasive strategy, findings were consistent across all prespecified subgroups based on physical frailty, comorbidity, and cognitive impairment. Adding such subgroup analyses helps to build the evidence base for clinical decision making when treating older adults. Pooling individual or group-level data in meta-analyses can further increase evidence on efficacy and tolerability. For example, a pooled analysis of five phase 3 trials from the Hellenic Oncology Research Group showed that adjuvant taxane-based therapy offered similar efficacy in older versus younger patients with breast cancer, but with increased toxicity in older patients.⁵⁵

Methodological challenges in clinical research involving older adults

Explanatory versus pragmatic trials

Research quality can be described through internal and external validity. Internal validity is increased if the study minimises bias. External validity reflects the degree to which the findings can be generalised or applied to routine clinical practice (figure). Explanatory trials test whether an intervention works under ideal, tightly controlled conditions. Their design intentionally maximises internal validity at the expense of generalisability.⁹⁷ Although this approach effectively reduces bias, the highly selected populations and artificial setting fail to reflect the reality of routine care. Consequently, real-world older populations are rarely adequately represented or considered in explanatory trials, and external validity is frequently compromised. For example, the registrational KEYNOTE-522 study⁹¹ for early triple-negative breast cancer successfully showed benefit of a new neoadjuvant immunotherapy-containing regimen; however, only 11% of the sample were aged at least 65 years at inclusion.

Pragmatic trials aim to evaluate whether interventions work under usual-care conditions, optimising external validity.⁹⁷ However, this design entails trade-offs that

threaten internal validity. Since treatment delivery, adherence, and clinical decisions vary naturally across settings, pragmatic trials are exposed to greater heterogeneity of patients and settings, implementation variability, and lower adherence, which can dilute treatment effectiveness. Given these trade-offs, investigators should use tools such as the Pragmatic Explanatory Continuum Indicator Summary⁹⁸ to explicitly justify and document the intended position of their trial along the explanatory–pragmatic continuum.

One obstacle in performing trials in older people that hampers external validity is the absence of gold standard definitions and measurement instruments for frailty. Different definitions of frailty can explore different domains (eg, medical, physical, dementia and depressive state, and social) and therefore studies on complex patients might be poorly comparable if the included patients are not well described.⁹⁹ Similarly, although the concept of resilience is attracting attention as it better aligns with patient goals than frailty, there is no consensus on its definition or measurements. Operationalisation of resilience in clinical trials or cohorts is therefore complicated.

Hybrid effectiveness–implementation trials

Traditionally, trials focus on the efficacy and tolerability of the treatment. The ability to translate the intervention into real-world practice is rarely, if ever, assessed. The gap between the intervention in a trial and real-world practice is often substantial and requires adjustments that can affect actual efficacy due to various human, economic, and logistical obstacles. These obstacles must be identified in advance, so that the intervention can subsequently be rolled out and achieve its intended impact. To this end, hybrid design tests have been developed.¹⁰⁰ Hybrid effectiveness–implementation trials aim to simultaneously evaluate effectiveness of a given intervention and its implementation in clinical care practice. Three main types have been described: type 1 hybrid trials focus primarily on effectiveness but include implementation endpoints as secondary (eg, feasibility, acceptability, and fidelity); in type 2, effectiveness and implementation are co-primary endpoints; and in type 3, implementation is the main endpoint and effectiveness is secondary. These trials could help accelerate the translation of evidence into practice.

Observational studies

Beyond RCTs, observational data sources play an increasingly important role in research on older adults. Because the populations captured in observational data are generally older, frailer, and display more comorbidity, they often provide a more realistic picture of the spectrum of patients treated in routine care than RCT populations. Some observational data sources are therefore known as real-world data and offer excellent generalisability. However, as with RCTs, the internal and

Data sources	Internal validity (risk of bias)		External validity (generalisability)		Examples
RCTs	High comparability between groups through randomisation		Explanatory: strict inclusion criteria, intervention control, and intensive monitoring	Pragmatic: inclusion criteria and intervention closer to usual-care conditions Hybrid (types 1–3): simultaneously test intervention effectiveness and implementation strategies	Pembrolizumab for early triple-negative breast cancer: KEYNOTE-522 (explanatory RCT); ⁹¹ invasive treatment strategy for older patients with myocardial infarction ⁵⁴
Claims databases	Few or some frailty, multimorbidity, or functional parameters		Near-exhaustivity of population coverage (county, region, or country)		Study on denosumab vs oral bisphosphonates among female patients who are dialysis dependent (used the Consolidated Renal Operations in a Web-Enabled Network database from Medicare and Medicaid Services) ⁹²
CDRs (eg, electronic health records)	Few or some frailty, multimorbidity, or functional parameters through medical notes		Monocentre	Multicentre	Study on nirmatrelvir or ritonavir vs molnupiravir in patients with COVID-19 who have comorbidities (used the Taipei Medical University Clinical Research Database; multicentre CDR) ⁹³
Epidemiological population registries (eg, cancer or stroke)	Scarce or no information on frailty, multimorbidity, or functional parameters		Near-exhaustivity of population coverage (county, region, or country)		Study on risk of stroke or systemic embolism and major bleeding associated with the use of oral anticoagulants in elderly patients with atrial fibrillation (used the Norwegian Patient Registry and the Norwegian Prescription Database) ⁹⁴
Clinical cohorts	Retrospective: scarce or no information on few frailty, multimorbidity, or functional parameters	Prospective: numerous frailty, multimorbidity, or functional parameters; residual confounding	Monocentre	Multicentre	Direct oral anticoagulants for atrial fibrillation in older adults and extremely older adults ⁹⁵ (prospective monocentre clinical cohort study); immune checkpoint inhibition and chemotherapy in older adults with non-small cell lung cancer ⁹⁶ (retrospective multicentre clinical cohort study)

Figure: Validity framework for research in older adults

Green represents a low risk of bias (internal validity) or high generalisability. Yellow represents some concerns with internal validity (mainly confounding bias) or generalisability. Red represents a high risk of bias (mainly confounding bias) or low generalisability. RCT=randomised controlled trial. CDR=clinical data repository.

external validity of observational studies, as well as their methodological rigour and interpretability, can differ substantially between studies. Four main types of observational data sources can be considered (figure): claims databases,⁹² which contain administrative data related to billing, reimbursement, diagnoses, procedures, and prescription dispensing; clinical data repositories (CDRs),⁹³ which aggregate electronic clinical information derived from structured (codes) and unstructured (medical notes) routine care (hospital and ambulatory); epidemiological population registries,¹⁰¹ which continuously and systematically record all cases of a disease occurring within a defined territory; and prospective⁹⁵ or retrospective⁹⁶ clinical cohorts, including baseline clinical and therapeutic information and longitudinal follow-up.

Notable differences persist between clinical cohorts, CDRs, registries, and claims databases. Single-centre cohorts, such as a prospective cohort study investigating direct oral anticoagulants in older patients with atrial fibrillation in Padova, Italy,⁹⁵ reflect only the characteristics of patients managed within a single institution, including its specific recruitment patterns,

socioeconomic context, and medical practices. In contrast, epidemiological registry data¹⁰¹ and health insurance claims⁹² usually cover an entire region or country and therefore offer far greater representativeness and generalisability. However, due to a failure to collect comorbidity and frailty parameters, internal validity remains low. Multicentre initiatives—including both tertiary and community-based centres—can enhance the external validity of cohorts and CDRs, as patient characteristics and care practices vary substantially across settings. However, many retrospective cohorts or CDRs contain no data on older people's characteristics, such as frailty and cognition, and lack outcomes relevant to older people such as PROs, limiting internal validity and confounding bias. From the standpoint of internal validity, prospective clinical cohorts allow for the systematic measurement of all known confounding factors and thus reduce the confounding bias better than all other observational data sources. However, the absence of randomisation prevents balancing of unmeasured confounders, resulting in residual confounding. Moreover, non-randomised studies can be subject to treatment selection bias, in which treated

patients differ systematically from non-treated patients.¹⁰² As a result, treatment effect cannot be estimated by simply comparing outcomes between treatment groups. Even multivariable models do not sufficiently address treatment-selection bias. Inverse probability of treatment weighting using the propensity score allows researchers to obtain less biased estimates of average treatment effects, as shown in a UK cohort study in 5238 women that linked two routine-care data sources to assess the effect of adjuvant trastuzumab-based treatment on survival according to patient age and fitness.¹⁰³

Moreover, treatment effect analyses in observational datasets are highly susceptible to immortal time bias, particularly when comparing an active treatment with no treatment. Treated patients are typically followed from the time of treatment initiation, whereas untreated patients are followed from diagnosis or hospital admission—often several days or weeks earlier. Patients who die early are therefore included only in the untreated group, which systematically biases results in favour of the treatment. This was clearly shown in a multicentre retrospective cohort study in UK on the benefit of geriatric assessment in adult trauma centres, in which Kaplan–Meier curves separate immediately.^{104,105} Addressing this issue through target trial emulation strategies, specifically designed to reduce the risk of immortal time bias, should therefore be considered.¹⁰³ Target trial emulation is a framework to assess the effect of treatment in observational studies by minimising the selection and confounding bias inherent in such designs. It works in two main ways: as a propensity score approach it helps to make the characteristics of the two treatment groups comparable, and thus, achieve a balance closer to the balance found in randomised trials; and by ensuring that the eligibility time, the start of follow-up, and treatment assignment are aligned (by using different methods such as grace period, cloning, weighting, and censoring) between the two groups.

Artificial intelligence (AI) in health is developing rapidly and could contribute to innovations in trial design and analyses.¹⁰⁶ Machine or deep learning algorithms could help substantiate the external validity of a trial, assist in adaptive trial design, and reduce confounding bias. In some cases, the concepts of digital twins or synthetic control arms could replace or augment standard RCTs. Methodological and ethical frameworks for AI are rapidly evolving in parallel with their implementation. One specific application is the use of generative AI to increase the use and validity of PROs.¹⁰⁷ Traditional PROs are measured as unidirectional questionnaires, whereas concepts like QoL are more complex. Generative AI could introduce conversational agents to increase patient engagement and follow-up over time, perform a more qualitative synthesis of verbal patient input beyond scores, and get a deeper understanding of concepts such as QoL by bottom-up

understanding of patient input as opposed to professional-designed questionnaires. Large language models might be very helpful for PROs, although their safety and validation need development. However, a concern around AI in health care is that it might perpetuate the biases in the data regarding health-care access; therefore, care must be taken to avoid ageism that could be perpetuated in AI-based work.

Conclusions

The older population continues to expand rapidly, forming a very large, diverse population with little trial evidence due to persistent obstacles to participation. Priorities for elevating trial applicability include recruiting a representative sample of older patients, selecting age-adapted interventions, and prioritising meaningful endpoints. Some all-age trials now better characterise the included population or include stratified interventions enabling real-world transposition. In older-age specific trials, there are now some excellent successful examples of trials that address relevant outcomes and interventions and are relevant to all older patients, including those with frailty. These trials can be complemented by alternative data sources and designs, such as various forms of observational studies based on different data sources including claims data and CDRs, which have their own advantages and limitations. However, although observational studies can complement the findings of randomised trials they should not be seen as an alternative to RCTs.

Regulatory agencies recognise the limited generalisability of registrational trials to older populations and have developed guidance for investigators in academia and industry. Both the European Medicines Agency and the FDA have requested sponsors to consider older people explicitly in trial design, provide data from subgroup analyses in older patients, remove restrictive eligibility criteria, and integrate geriatric-relevant endpoints.^{108,109} For most agencies, the guidance is non-binding, which limits uptake. Much information on older people can also be retrieved from registrational trials on new drugs that are increasingly coordinated by the pharmaceutical industry. Age subanalyses can provide important information for drug efficacy as well as toxicity according to age and frailty status but are not systematically performed or are suboptimally performed. It is important that registrational trial data are shared with academic researchers,¹¹⁰ allowing more accurate identification of the population who benefits or does not benefit from new treatments, since this benefit can be different for older versus younger people.

To ensure trial design and conduct that aligns with the specific needs and characteristics of older people, patient and public involvement is recommended. Patients can actively contribute as listeners, advisors, or decision makers.¹¹¹ Patient and public involvement in cardiovascular clinical trials has been advocated by the Global

Cardiovascular Clinical Trialists Workshop, with patients involved throughout the drug development lifecycle from pretrial planning to postapproval clinical use.¹¹² There are still many barriers to overcome in clinical trial design for older adults, but there have been major improvements and many examples of successful trials in the previous decades. Researchers should carefully address these barriers and solutions in trial design to ensure inclusion of older and frail populations in future clinical trials to improve external validity.

Contributors

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Declaration of interests

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